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General Nonsymmetric Higher-Order Decomposition of Exponential Operators and Symplectic Integrators

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A general scheme to realize nonsymmetric higher-order decomposition of exponential operators and symplectic integrators is constructed. This gives a unified theory of the previous symmetric and nonsymmetric decomposition. Ruth's nonsymmetric third-order real decomposition is shown to be a simple example of the present general theory.

As discussed in previous papers,[1-3] it is possible to realize recursively higher-order decomposition of exponential operators and symplectic integrators up to an infinite order. Even fourth-order or sixth-order decomposition is very useful in particle accelerator physics,[4-7] molecular physics,[8] astrophysics[9] and statistical physics.[10]

The present problem is to find the following product approximant

$$e^{x(A+B)}=e^{t_1xA}e^{t_2xB}e^{t_3xA}e^{t_4xB}\cdots e^{t_MxA}+O(x^{m+1}), \tag{1}$$

with real numbers $\{t_j\}$ and with finite M for any noncommutable operators A and B. In statistical physics, the exponential operator $\exp[x(A+B)]$ denotes the density matrix $\exp[-\beta\mathscr{H}]$ with the Hamiltonian $\mathscr{H}$ and the inverse temperature $\beta=1/k_BT$. In quantum mechanics, the parameter x in eq. (1) denotes imaginary time, namely $x=it/\hbar$ with Planck's constant $\hbar$. In nonlinear physics, the operators A and B denote symplectic integrators and the variable x denotes real time t. In this case, the operators A and B are nonlinear.

Here, we are interested in the algebraic structure of general decomposition of eq. (1) and consequently we assume here that the operators A and B are arbitrary. Thus, our scheme of decomposition is quite general and it is applicable to any problem in which the relevant solution $f(x)$ is expressed in the form

$$f(x)=e^{x(A+B)}f(0). \tag{2}$$

The first-order decomposition of the exponential operator $\exp[x(A_1+A_2+\cdots+A_q)]$ for any noncommutable operators $\{A_j\}$ is given by

$$F_1(x)=e^{xA_1}e^{xA_2}\cdots e^{xA_{q-1}}e^{xA_q}, \tag{3}$$

as is well known. Of course, there are many similar expressions such as

$$\tilde{F}_1(x)=e^{xA_q}e^{xA_{q-1}}\cdots e^{xA_2}e^{xA_1}. \tag{4}$$

The second-order decomposition may be given by the following symmetrized decomposition[11] $S(x)=F_1(x/2)\tilde{F}_1(x/2)$. This has the following symmetry property: $S^{-1}(x)=S(-x)$ or $S(x)S(-x)=1$. In the previous papers,[1-3] the present author proposed the following recursive scheme: The m-th order decomposition $Q_m(x)$ is constructed as

$$Q_m(x)=Q_{m-1}(p_{m1}x)Q_{m-1}(p_{m2}x)\cdots \times Q_{m-1}(p_{mr}x), \tag{5}$$

with the condition that

$$\sum_{j=1}^{r}p_{mj}^m=0 \text{ and } \sum_{j=1}^{r}p_{mj}=1. \tag{6}$$

It should be noted here that there always exists a real solution p_{mj} for odd integer m but that the solution of eq. (6) becomes complex for an even integer m. However, if we start from the second-order symmetric decomposition $S(x)$ in (5) with the symmetry condition $p_{m(r-j+1)}=p_{mj}$ for all j, then the odd-order decomposition $Q_{2m-1}(x)$ becomes symmetric and consequently it is correct[1-3,11] up to the $2m$-th

order. Thus, we have realized any higher-order real decomposition recursively even up to an infinite order.

Then arises the question of whether or not we can realize general nonsymmetric higher-order decomposition systematically. This question is answered here affirmatively. This scheme of nonsymmetric decomposition is practically very important, because this nonsymmetric scheme makes it much easier to realize fractal decomposition with rational parameters, as shown in the text. We first give the following general scheme of decomposition. The general m-th order (not necessarily symmetric) decomposition $F_m(x)$ is shown as a product of several (not necessarily the same) approximants $G_1(x)$, $G_2(x), \cdots$, $G_r(x)$, namely

$$F_m(x)=G_1(p_1x)G_2(p_2x)\cdots G_r(p_rx), \quad (7)$$

with the trivial condition $p_1+p_2+\cdots+p_r=1$, as before.[1-3] Here, we assume that the approximant $G_j(x)$ is of the order of s_j. (The parameter s_j denotes a positive integer.) Then, it is expressed as

$$G_j(x)=\exp\,(x\mathscr{H}+x^2R_{j2}+x^3R_{j3}+\cdots+x^mR_{jm}+\cdots), \quad (8)$$

with $\mathscr{H}=A_1+A_2+\cdots+A_q$ and with some appropriate operators $\{R_{jm}\}$ which are composed of linear combinations of commutators of $\{A_j\}$. The assumption that $G_j(x)$ is of the order of s_j requires that

$$R_{j2}=R_{j3}=\cdots=R_{js_j}=0 \text{ and } R_{j(s_j+1)}\neq 0. \quad (9)$$

Here we introduce, in general, the tilde operator $\tilde{G}(x)$ by the inverse-order product of $G(x)$, as in eq. (4). It has the property $\tilde{G}(x)=G^{-1}(-x)$. Furthermore, the symmetry of $G(x)$ is defined by $\tilde{G}(x)=G(x)$ namely $G(x)\times G(-x)=1$. In a similar manner to eq. (9), we may write

$$\tilde{G}_j(x)=\exp\,(x\mathscr{H}+x^2\hat{R}_{j2}+x^3\hat{R}_{j3}+\cdots+x^m\hat{R}_{jm}(x)+\cdots). \quad (10)$$

Then, we find that $\hat{R}_{j(2m-1)}=R_{j(2m-1)}$ and $\hat{R}_{j(2m)}=-R_{j(2m)}$, using the above property. These expressions will be used effectively later to obtain systematic nonsymmetric higher-order decomposition.

Now we write the m-th order product $F_m(x)$ in the following form:

$$\begin{aligned}F_m(x)&=\prod_{j=1}^{r}G_j(p_jx)=\prod_{j=1}^{r}\exp\,\{(xp_j)\mathscr{H}+(xp_j)^2R_{j2}+(p_jx)^3R_{j3}+\cdots\}\\&=P\exp\left[\sum_{j=1}^{r}\{(xp_j)\mathscr{H}+(xp_j)^2R_{j2}+(xp_j)^3R_{j3}+\cdots\}\right].\end{aligned} \quad (11)$$

Here the symbol P denotes the "time-ordering" operation with respect to the subscript "j", as used in refs. 2 and 3. Furthermore we make use of Kubo's symmetrization operation[12] S with respect to the same subscript "j". It is defined by

$$S(R_{jm}^pR_{jn}^q\cdots)=\frac{p!q!\cdots}{(p+q+\cdots)!}\sum \text{Permu}\,(R_{jm}^pR_{jn}^q\cdots), \quad (12)$$

for any positive integers $p, q, \cdots$. The symbol "Σ Permu" in eq. (12) denotes the summation of all the operators obtained in all possible ways of permutation. Then the m-th order decomposition $F_m(x)$ is expressed as

$$\begin{aligned}F_m(x)&=P\exp\,(xY_1+x^2Y_2+x^3Y_3+\cdots)\\&=PS(e^{xY_1}e^{x^2Y_2}e^{x^3Y_3}\cdots)\\&=\sum_{n_1,n_2,n_3,\cdots}\frac{x^{n_1+2n_2+3n_3+\cdots}}{n_1!n_2!n_3!\cdots}PS(Y_1^{n_1}Y_2^{n_2}Y_3^{n_3}\cdots),\end{aligned} \quad (13)$$

where

$$Y_1=\sum_{j=1}^{r}(p_j\mathscr{H}) \quad \text{and} \quad Y_n=\sum_{j=1}^{r}(p_j^nR_{jn}), \quad (14)$$

for $n=2, 3, \cdots$. Expression (13) is our general basic formula of the m-th approximant containing the parameters $\{p_j\}$. In the above general expression of eqs. (13) with eqs. (14), the parameter p_j should not be separated from $\mathscr{H}$, nor p_j^n from R_{jn}, before the operations P and S are performed, particularly when R_{jn} is independent of the subscript "j", as is seen later. This is because the operators $\mathscr{H}$ and $R_n(=R_{jn})$ have to be ordered through the operations P and S in terms of the parameters $\{p_j\}$ as

$$P\{(p_j\mathscr{H})(p_k^n R_n)\}=\begin{cases}(p_j\mathscr{H})(p_k^n R_n)=(p_j p_k^n)\mathscr{H} R_n & \text{for } j<k\\ (p_k^n R_n)(p_j\mathscr{H})=(p_j p_k^n)R_n\mathscr{H} & \text{for } j>k,\end{cases}$$

and

$$S\{(p_j\mathscr{H})(p_j^n R_n)\}=p_j^{n+1}\frac{1}{2}(\mathscr{H}R_n+R_n\mathscr{H}).$$

Now, eq. (13) is easily rewritten as

$$F_m(x)=e^{x\mathscr{H}}+\sum_{n_1,n_2,n_3,\cdots}{}' \frac{x^{n_1+2n_2+3n_3+\cdots}}{n_1!\,n_2!\,n_3!\cdots}PS(Y_1^{n_1}Y_2^{n_2}Y_3^{n_3}\cdots), \tag{15}$$

with $\mathscr{H}=A_1+A_2+\cdots+A_q$, where the symbol Σ' in eq. (15) denotes the summation over $n_1, n_2, n_3, \cdots$ excluding $n_2=n_3=\cdots=0$. Thus, the condition for $F_m(x)$ to be of the m-th order is reduced to the conditions that

$$PS(Y_1^{n_1}Y_2^{n_2}Y_3^{n_3}\cdots)=0, \tag{16}$$

for all nonnegative integers $n_1, n_2, \cdots$ under the restriction that $n_1+2n_2+\cdots\leq m$, excluding $n_2=n_3=\cdots=0$, where Y_n are defined by eq. (14) in terms of the parameters $\{p_j\}$. Therefore these parameters $\{p_j\}$ are determined from the conditions in eq. (16).

The above conditions are very abstract, but they are extremely convenient for determining many practical schemes of higher-order decomposition, as will be discussed later. In fact, they can be expressed explicitly using simple procedures, at least for lower-order decomposition.

There are some interesting schemes in which equations used to determine the parameters $\{p_j\}$ can be obtained explicitly from the general conditions in eq. (16).

i) Nonsymmetric complex decomposition

We choose $\{G_j(x)\}$ in eq. (7) as $G_1(x)=G_2(x)=\cdots=G_r(x)=F_1(x)$. Then, the operator Y_n in eq. (14) is reduced to

$$Y_n=\sum_{j=1}^{r}(p_j^n R_n), \tag{17}$$

where $R_{1n}=R_{2n}=\cdots=R_{rn}\equiv R_n$. This gives the general complex decomposition described in the previous paper,[3] because the conditions $PS(Y_n)=0$ for $n=2, 3, \cdots m$ give the equations that $p_1^n+p_2^n+\cdots p_r^n=0$ for $n=2, 3, \cdots, m$.

ii) Recursive scheme

We choose $\{G_j(x)\}$ as $G_1(x)=G_2(x)=\cdots=G_r(x)=Q_{m-s}(x)$ with the $(m-s)$-th order decomposition $Q_{m-s}(x)$. Then we obtain the following recursive scheme $Q_m(x)=Q_{m-s}(p_1x)Q_{m-s}(p_2x)\cdots Q_{m-s}(p_rx)$. The parameters $\{p_j\}$ are determined from the conditions in eq. (16) under the restriction that $m+1-s\leq n_1+2n_2+3n_3+\cdots+mn_m\leq m$ with Y_n defined by eq. (17) for $m+1-s\leq n\leq m$ and with $Y_2=Y_3=\cdots=Y_{m-s}=0$. In particular, the case $s=1$ corresponds to the recursive decomposition of eqs. (5) with eq. (6).

iii) Symmetric decomposition

We choose $\{G_j(x)\}$ as $G_1(x)=G_2(x)=\cdots=G_r(x)=S(x)$ with the second-order symmetric decomposition $S(x)$ defined by $S(x)=F_1(x/2)\tilde{F}_1(x/2)$, and with the symmetric parameters $\{p_j\}$ satisfying the relationship $p_{r+1-j}=p_j$. Then we obtain a $(2m-1)$-th order symmetric decomposition of the form $F_{2m-1}(x)=S(p_1x)S(p_2x)\cdots S(p_rx)$ with $p_{r+1-j}=p_j$, as shown in ref. 3.

iv) Nonsymmetric real decomposition

If we choose $\{G_j(x)\}$ as $G_1(x)=G_3(x)=\cdots=F_1(x)$ and $G_2(x)=G_4(x)=\cdots=F_1(x)$, namely

$$F_m(x)=F_1(p_1x)\tilde{F}_1(p_2x)F_1(p_3x)\tilde{F}_1(p_4x)\cdots \times F_1(p_{r-1}x)\tilde{F}_1(p_rx), \tag{18}$$

then the operators $\{Y_n\}$ in eq. (14) are expressed as

$$Y_{2n}=\sum_{j=1}^{r}(-1)^{j-1}p_j^{2n}R_{2n},$$

and

$$Y_{2n-1}=\sum_{j=1}^{r}p_j^{2n-1}R_{2n-1}, \tag{19}$$

with eq. (17). Thus, for example, the conditions $\mathrm{PS}(Y_n)=0$ for $n=2, 3,\cdots, m$ give the relationships $p_1^{2n}+p_3^{2n}+p_5^{2n}+\cdots=p_2^{2n}+p_4^{2n}+p_6^{2n}+\cdots$ for $2n\leq m$ and $p_1^{2n-1}+p_1^{2n-1}+\cdots+p_r^{2n-1}=0$ for $3\leq 2n-1\leq m$. Obviously there exist real solutions of $\{p_j\}$ for r large enough. The above general nonsymmetric decomposition is a new scheme in the present paper. Thus, we discuss here in greater detail some explicit lower-order decomposition schemes.

First we discuss the second-order nonsymmetric decomposition in the form

$$F_2(x)=F_1(p_1x)\tilde{F}_1(p_2x)F_1(p_3x)\tilde{F}_1(p_4x)\cdots \times F_1(p_rx). \tag{20}$$

The parameters $\{p_j\}$ are determined by the condition $PS(Y_2)=0$, namely $p_1^2+p_3^2+\cdots=p_2^2+p_4^2+\cdots$ as well as the normalization $p_1+p_2+p_3+\cdots=1$. In the case of $r=2$, we obtain $p_1+p_2=1$ and $p_1^2=p_2^2$. This yields $p_1=p_2=1/2$ and consequently the symmetric decomposition $S(x)=F_1(x/2)\tilde{F}_1(x/2)$. In the case of $r=3$, we obtain the equations $p_1+p_2+p_3=1$ and $p_1^2+p_3^2=p_2^2$. There are an infinite number of real solutions. In particular, we obtain the following rational solution $p_1=(m-n)/(2m)$, $p_2=(m^2+n^2)/(2m(m+n))$, and $p_3=n/(m+n)$ for any integers m and n satisfying the condition $m(m+n)\neq 0$. For example, we have $p_1=1/4$, $p_2=5/12$, $p_3=1/3$ namely $F_2(x)=\mathrm{e}^{(1/4)xA}\,\mathrm{e}^{(2/3)xB}\mathrm{e}^{(3/4)xA}\,\mathrm{e}^{(1/3)xB}$. Moreover, other mixed products such as $F_1'(x)=\mathrm{e}^{t_1A_1}\mathrm{e}^{t_2A_2}\mathrm{e}^{t_3A_1}\mathrm{e}^{t_4A_2}\mathrm{e}^{t_5A_3}\cdots\mathrm{e}^{t_rA_q}$ may be used instead of $F_1(x)$ in our general scheme, if it is convenient.

Next, we discuss the third-order decomposition $F_3(x)$ in this scheme of nonsymmetric decomposition. The conditions in eq. (16) are given by $PS(Y_2)=PS(Y_3)=PS(Y_1Y_2)=0$ in this case. They are explicitly rewritten as

$$p_1^2+p_3^2+\cdots=p_2^2+p_4^2+\cdots,$$
$$p_1^3+p_2^3+p_3^3+\cdots+p_r^3=0, \tag{21}$$

and

$$\sum_{k=1}^{r}p_k\hat{a}_{2k}=0, \tag{22}$$

with

$$\hat{a}_{2k}=\sum_{j<k}(-1)^{j-1}p_j^2+\frac{1}{2}(-1)^{k-1}p_k^2. \tag{23}$$

Note that $\hat{a}_{21}=(1/2)p_1^2$.

In the case of $r\leq 4$ there is no real solution $\{p_j\}$ to eqs. (21) and (22) together with $p_1+p_2+\cdots+p_r=1$. In the case of $r=5$, eqs. (21) and (22) are expressed as

$$p_1^2+p_3^2+p_5^2=p_2^2+p_4^2,$$
$$p_1^3+p_2^3+p_3^3+p_4^3+p_5^3=0, \tag{24}$$

and

$$f(p_1, p_2, p_3)=f(p_5, p_4, p_3), \tag{25}$$

with $f(x, y, z)=x^2(1-x)+y^2(2x+y-1)-z^2(x+y)$. If we assume that $p_2=p_3$, then we find the following rational solution $p_1=7/24$, $p_2=p_3=3/8$, $p_4=-25/24$, $p_5=1$. The third-order decomposition $F_3^{(R)}(x)$ for this set of parameters is rewritten as $F_3^{(R)}(x)=\mathrm{e}^{(7/24)xA}\mathrm{e}^{(2/3)xB}\mathrm{e}^{(3/4)xA}\mathrm{e}^{-(2/3)xB}\mathrm{e}^{-(1/24)xA}\mathrm{e}^{xB}$ for the case $\mathscr{H}=A+B$. This was found first by Ruth.[4] The present third-order nonsymmetric decomposition $F_3(x)=\mathrm{e}^{p_1A_1}\mathrm{e}^{p_1A_2}\cdots\mathrm{e}^{(p_1+p_2)A_q}\mathrm{e}^{p_2A_{q-1}}\cdots\mathrm{e}^{(p_2+p_3)A_1}\times\mathrm{e}^{p_3A_2}\cdots\times\mathrm{e}^{(p_4+p_5)A_1}\mathrm{e}^{p_5A_2}\cdots\mathrm{e}^{p_5A_q}$ with the above rational integers $\{p_j\}$ is a generalization of Ruth's formula for $\mathscr{H}=A+B$ for the case $\mathscr{H}=A_1+A_2+\cdots+A_q$. There are an infinite number of other real solutions $\{p_j\}$ to eqs. (24) and (25) together with $p_1+p_2+p_3+p_4+p_5=1$. For example, if we assume that $p_1=p_5$ and $p_2=p_4$, then the condition in eq. (25) is automatically satisfied. Since eqs. (24) and (25) are homogeneous equations, we obtain $12u^4-8u^3+6u^2-1=0$ for the ratio $u=p_1/p_3$, while the ratio $v=p_2/p_3$ is given by $v^2=u^2+1/2$. The equation $p_1+\cdots+p_5=1$ determines the normalization of the parameters $\{p_j\}$. Thus we find that there are

the following two sets of real solutions:
Solution A:

$$p_1=p_5=0.2683300957817599\cdots,$$
$$p_2=p_4=0.6513314272356399\cdots,$$
$$p_3=-0.8393230460347997\cdots.$$

Solution B:

$$p_1=p_5=1.995,$$
$$p_2=p_4=-3.690,$$
$$p_3=4.390.$$

It should be noted that the parameters $\{p_j\}$ of Solution A satisfy the inequality $|p_j|<1$ for all j's. This property is very important in practical applications, because the convergence of the decomposition improves in the situation where $|p_j|<1$ for $j=1, 2, \cdots$, and r. From this point of view, Solution B is not practical.

Ruth's formula $F_3^{(R)}(x)$ is particularly interesting in the sense that the parameters $\{p_j\}$ are rational numbers, because they are very convenient in practical applications. The fourth-order decomposition with rational numbers $\{p_j\}$ is given by

$$F_4(x)=F_3^{(R)}\left(\frac{x}{2}\right)\tilde{F}_3^{(R)}\left(\frac{x}{2}\right)=e^{\frac{7}{48}xA}e^{\frac{1}{3}xB}e^{\frac{3}{8}xA}e^{-\frac{1}{3}xB}e^{-\frac{1}{48}xA}e^{xB}e^{-\frac{1}{48}xA}e^{-\frac{1}{3}xB}e^{\frac{3}{8}xA}e^{\frac{1}{3}xB}e^{\frac{7}{48}xA},$$

for the case $\mathscr{H}=A+B$.

In a similar way, it is easy to obtain a nonsymmetric higher-order decomposition $F_{2m-1}(x)$ explicitly in the above scheme. Then, the $2m$-th order symmetric decomposition $F_{2m}(x)$ can be obtained as $F_{2m}(x)=F_{2m-1}(x/2)\tilde{F}_{2m-1}(x/2)$.

More explicit estimates of the parameters $\{p_j\}$ for some higher-order decomposition in this new nonsymmetric decomposition will be published very soon elsewhere, together with mathematical analyses of the structure of $PS(Y_1^{n_1}Y_2^{n_2}Y_3^{n_3}\cdots)$, using free Lie algebra, particularly Witt's formula and the Möbius function.[3)]

v) Recursive nonsymmetric real decomposition

We consider the following recursive scheme: $F_m(x)=F_{m-s}(p_{m1}x)\tilde{F}_{m-s}(p_{m2}x)\cdots F_{m-s}(p_{mr}x)$. The parameters $\{p_j\}$ are determined by the conditions in eq. (16) with Y_n defined by eq. (19).

For example, in the case that $s=1$ and $r=3$ for even m, we have the solution $p_{m1}=p_{m3}=(2+2^{1/m})^{-1}$ and $p_{m2}=2^{1/m}/(2+2^{1/m})$. It should be noted that $|p_{mj}|<1$ for all j and even m. For odd m, we have the ordinary condition that $p_{m1}^m+p_{m2}^m+\cdots+p_{mr}^m=0$.

Finally it should be emphasized that the present scheme yields a unified theory of symmetric and nonsymmetric higher-order decomposition of exponential operators and symplectic integrators.

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